‘An Explanation of Short-run changes in U.K. Inflation, 2000 – 2025’.

1. INTRODUCTION

Understanding UK inflation attentively has become a challenge over the past twenty-five years, as the UK economy experienced macroeconomic changes from 2000 to 2025 such as global monetary crisis, Brexit vote, COVID-19 and due to this inflation went up and down. These fluctuations have made it important to understand what influences inflation to change in both long run and the short run.

This report aims to explain what factors influences U.K. inflation to change in the short run, using monthly data from 2000 to 2025. It focuses on three key economic variables, which are the BOE interest rate, the unemployment rate, and Brent crude oil prices. These variables were chosen to represent most of the important parts of the economylike monetary policy, how many people are out of work, and how expensive the price of energy is. All of these can influence how rapidly prices rise in the U.K.

These relationships are examined using Johansen cointegration analysis and a Vector error Correction Model (VECM). These techniques help in studying the long-run equilibrium behavior and short-run adjustments. All the empirical calculations, statistical tests, and estimations of models were conducted using Stata. The main findings that come out are:

- The Johansen test shows that inflation and other three variables share one cointegrating relationship.
- The Brent crude oil price is the only significant long run determinant of U.K. inflation.
- In the Short run, inflation adjusts slowly about 4.88% per month towards its long-run equilibrium.
- Lagged inflation and BOE interest-rate movements significantly influence short-run inflation.

The rest of the report summarizes the theoretical and empirical literature, describes the data and methodology, Presents the empirical results, and gives their interpretation and concludes with limitations and policy recommendations pointing out the bibliography.

2. BRIEF LITERATURE REVIEW

A usual starting point for inflation modelling is via the Phillips Curve, which suggests a negative relation between unemployment and inflation (Phillips, 1958). However, the modern UK Phillips Curve is widely regarded as having lost much of its previous slope (Broadbent,2014), which means that unemployment has less explanatory power for inflation.

Monetary policy is the key influencing factor of U.K. inflation. Since 1997, the BOE has relied on changes in interest rates to meet the government's inflation target. It’s said that the

monetary policy works with long and variable lags, which means changes in interest rate influence inflation gradually over several months (Batini and Nelson, 2001).

Energy prices, and in particular oil and gas have been recognized as prominent contributors to U.K. inflation. As the U.K. depends on imported energy, Brent crude oil price shocks have been transmitted into domestic inflation (Beechey & Osterholm, 2008).

Cointegration is widely used in the literature as source of investigating long-run macroeconomics relationships. The theoretical groundwork is given by Johansen (1991) and Engle & Granger (1987), and later studies used it to look specifically at the U.K. economy.

3. DATA AND METHODOLOGY

3.1 Data Source

The dataset consists of monthly observation from 2000m1 to 2025m9, which are taken from:

1. Inflation (CPI annual rate) from office for National Statistics (ONS)
2. Bank of England base rate from Bloomberg (UKBRBASE Index)
3. Unemployment rate of U.K. from Bloomberg (UKUEILOR Index)
4. Brent crude oil price from Bloomberg (CO1 Comdty)

The oil price variable was transformed using the natural logarithm (ln) – l_Cop = ln (Cop) because of its high volatility and right skewness.

3.2 Time -Series Properties

Time-series plots for the variables of inflation, interest rate, unemployment, and oil prices show clear trends as well as structural movements over time, which support the relevance of unit-root testing in confirming that the variables are likely to be non-stationary in levels.

Since the dataset consists of monthly macroeconomics variables from 2000 to 2025, the series must be checked for non-stationarity. Many time-series variables normally have trends, and running regressions in any non-stationary data often yields spurious results (Granger &Newbold,1974). It is necessary that each variable be checked for unit roots. When there are multiple variables that are integrated of order, I (1), they can still be in a stable long-run equilibrium – a concept called cointegration (Engle & Granger, 1987). The appropriate modeling framework in this case is the VECM, which combines thelong run cointegration relationship with short run adjustment dynamics (Johansen,1991).

3.3 Methodological Approach

1. Unit Root Testing
   The Augmented Dickey-Fuller test is used to evaluate whether the variables are stationary or non-stationary.

2. Lag-Length Selection
   A VAR length selection approach was conducted to determine the optimal lag using the AIC, HQIC, and SBIC measures. Where HQIC and SBIC suggested concise models such as lag 2 or lag 4, but AIC, FPE, and the log-likelihood all indicated that the model with 7 lags fits better. The seven lag model specification yielded the robust sample fit and produced good diagnostic result with full system stability unlike lag 2 or lag 4. For these reasons, lag 7 chosen as the final lag length for the analysis.

3. Cointegration Testing
   The Johansen Trace test identified One cointegrated vector, indicating a long-run equilibrium between the variables.

4. Estimating of Vector Error Correction Model
   In estimating the VECM, rank 1 was specified along with lag 7, as indicated by Varsoc, which enables the separate distinction of Long-run relationships, where it is captured by the cointegration equation and the short-term adjustments, which are represented by differenced variables and the Error-correction term.

5. Model diagnostic checks.
   Residual autocorrelation tests Veclmar and eigenvalue stability checks were executed to make sure the model is valid and there is no serial correlation.

4. RESULTS AND INTERPRETATION

1. Stationarity results
   All variables were non-stationary in their levels but became stationary after first difference, where the p value is < 0.05. This suggests that each series is integrated at order one I (1). Since all variables possess the same order of integration, a long-run relationship between them can be meaningfully evaluated.

2. Cointegration Test Results
   The Johansen cointegration test identifies one cointegration equation that exists in the long-run. Inflation, BOE base rate, unemployment and oil prices all move in a stable long run equilibrium. By establishing single cointegration relationships, this supports the use of an

error correction model, which allows for the modelling of the long-run equilibrium behavior along with the adjustment dynamics in the short run.

3. Long-Run Relationship

The long-run equilibrium equation, normalized on inflation, is:

ECT_t = inf_t - 0.1161 intrate_t + 0.0098 Unemp_t - 2.1393 ln(oil_t) + 6.4692

- Interest Rate: The long-run interest rate coefficient is statistically insignificant(p=0.328), indicating that the BOE Base rate does not directly determine long-run inflation.
- Unemployment Rate: In addition, unemployment is insignificant (p=0.960) in the long-run, consistent with the evidence of the relative flatness of the modern U.K. Phillips Curve and a weak long-run relationship between unemployment and inflation.
- Brent Crude oil prices: The coefficient of the log oil price is large and statistically significant. Oil prices shock raises long-run inflation because domestic prices are strongly influenced by developments in world energy cost. This justifies the fact that U.K. is a net importer of energy.

4. Short-Run Dynamics

- The Error correction term has a coefficient = -0.0487869, p = 0.001

This negative and statistically significant coefficient suggests that around 4.88% of any deviation from the long-run equilibrium is corrected every month. While adjustment is slow but significant.

The inflation dynamic in the short-run is summarized by the following compact error-correction model:

$$\Delta inf_t = -0.0487869 \cdot ECT_{t-1} + \Sigma_{i=1}^{6} \gamma_i \cdot \Delta X_{t-i} - 0.0017909 + \varepsilon_t$$

In the short run, lagged inflation terms at lags 3,4 and 6 are significant, which illustrates the strong persistence of inflation. Some lagged interest rate terms are significant, suggesting that monetary policy affects inflation after a while, rather than immediately. Lagged unemployment terms are significant as well, showing its short-run effect. Short-run contribution is also made by oil prices, whose impact is particularly strong in long-run equilibrium.

5. Model Diagnostic

Diagnostic tests confirm that the model is well specified as there is no residual autocorrelation (LM test), justifying that the chosen lag structure is appropriate. All eigenvalues lie inside the unit circle demonstrating that the VECM is stable. The inflation equation for R-squared is 0.3296, which is reasonable for monthly macroeconomics observations where random shocks and volatility are common.

5. CONCLUSION

The analysis performed here examined those factors determining UK short-run and long-run inflation from 2000–2025. The result points to one binding long-run equilibrium relationship among inflation, interest rates, unemployment, and Brent crude oil prices; there, only the log of oil prices exerts a statistically significant long-run impact. In the short run, inflation converges to its equilibrium rather slowly, in fact correcting about 4.88% of any disequilibrium in each period. The strong findings indicate how external energy costs remain a leading determinant of UK inflation, where domestic monetary policy factors have an impact only with a considerable lag in the short run.

6. LIMITATION AND POLICY RECOMMENDATIONS

Limitation:

- The model excludes structural breaks including Brexit and Covid-19.
- Unavailability of core inflation data.
- Only three explanatory variables were used; more could be added.

Policy Recommendations:

- It reduces dependence on imported energy. Being that the oil prices strongly influence inflation, it is urgent that the U.K. turns to renewable energies and increases energy independence.
- Enhancing monetary policy communication since the effects of interest rates seem to appear in the short term, and therefore clear communication will be helpful.
- It is better to enhance inflation forecasting models.

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